

CS 255 Business Requirements Document Template

Complete this template by replacing the bracketed text with the relevant information.

This template lays out all the different sections that you need to complete for Project One. Each section has guiding questions to prompt your thinking. These questions are meant to guide your initial responses to each area. You are encouraged to go beyond these questions using what you have learned in your readings. You will need to continually reference the interview transcript as you work to make sure that you are addressing your client's needs. There is no required length for the final document. Instead, the goal is to complete each section based on your client's needs.

Tip: You should respond in a bulleted list for each section. This will make your thoughts easier to reference when you move into the design phase for Project Two. One starter bullet has been provided for you in each section, but you will need to add more.

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- The purpose of the project is to develop a comprehensive driver training and management platform that helps students prepare for their DMV driving tests through online coursework, practice exams, and in-person driving lessons.
- The client is DriverPass and they want the system to allow customers to create accounts, schedule driving lessons, track progress, and access learning materials online. It should also let employees manage customer accounts, appointments, reports, and lesson information.

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- DriverPass wants the system to allow students to schedule driving lessons with instructors and assigned vehicles. It should also allow remote access to company data while keeping accurate scheduling and reservation tracking. Also, it should track relationships between students, vehicles, instructors, and appointment times to avoid scheduling conflicts.
- The problem DriverPass wants to fix is that many individuals fail DMV driving exams because of their limited access to effective training resources and updated study materials.
- The different components they would need would be:
 - Web-Based Cloud Interface: Which would provide secure access for students, instructors, administrators, and office staff.
 - Reservation and Scheduling System: That would manage lesson scheduling, instructor assignments, vehicle availability, and appointment modifications.
 - Learning Management Module: To allow management to generate downloadable reports and export data to Excel.
 - DMV Update Integration: To receive notifications and updates related to DMV policies, rules, and practice test changes.

- User Management System: Used to support role-based access and control account administration.

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

Upon completion it should:

- Enable students to register for accounts and manage appointments online.
- Allow customers to schedule, modify, and cancel two-hour driving lessons through the web interface.
- Prevent double-booking of instructors, vehicles, or lesson times.
- Provide students with progress tracking tools that display lesson completion, practice test scores, and exam status.
- Maintain accurate records of customer appointments, lesson notes, and account activity.
- Support role-based permission for administrators, IT staff, instructors, secretaries, and students.
- Implement a notification system that alerts the business when DMV policies or testing requirements change.
- Generate reports that can be downloaded for management review and offline analysis.
- Provide a responsive interface that works across desktop and mobile platforms.

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- The system must operate as a web-based application accessible through desktop and mobile browsers.
- The system should provide fast response times and reliable access for anyone accessing them. It should also allow online access to customer data from different locations.
- The system should receive updates whenever the DMV rules or practice test information changes.

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- The system should run on commonly used web browsers.

- The backend should have a centralized database is required to store customer records, appointments, instructor assignments, lesson history, and payment information. It must also support exporting reports and data in formats compatible with Excel.

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- The system would use role-based access control to assign permissions based on whether the user is an administrator, IT staff member, secretary, instructor, or student.
- Input fields should validate customer and scheduling information before the data is stored in the database.
- Administrators should receive notifications when scheduling conflicts, failed login attempts, or when system errors happen.
- Audit logs should also accurately track who created, modified, or canceled appointments.

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- Administrators and IT staff should be able to add, modify, disable, or remove user accounts without changing the system/s code.
- The cloud-based design should allow the platform to remain compatible with future browser and operating system updates.
- IT administrators should have full system access to manage accounts, reset passwords, and disable employee access when necessary.

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a "brute force" hacking attempt? What happens if the user forgets their password?

- Users must log in using secure account credentials such as username and password.
- Sensitive customer information and payment data should be encrypted during transmission and storage. The system should use secure HTTPS connections between the client and server.
- User accounts should temporarily lock after multiple failed login attempts to reduce brute-force hacking attempts.
- If a user forget their password, they should be able to complete a secure password recovery process.

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."

- The system shall allow students to schedule, modify, and cancel driving appointments online.
- The system shall assign instructors and vehicles to scheduled lessons.
- The system shall track customer progress for online practice tests and driving lessons.
- The system shall allow instructors to enter lesson notes and comments.
- The system shall notify administrators when DMV policies or practice test information changes.
- The system shall generate downloadable reports for management review.

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- The interface should support students, instructors, office staff, administrators, and IT staff.
- Students should be able to access online classes, practice exams, lesson schedules, and progress reports. Secretaries should be able to schedule appointments and manage customer accounts. Administrators and IT should have access to reporting, account management, and security settings.
- The interface should include responsive web design so users can access the system from desktop, and mobile browsers.
- Scheduling calendars, dashboards, and input forms should be simple and easy to navigate.

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- The design assumes users will have reliable internet access for scheduling and online learning.
- The system assumes customers will access the platform using modern web browsers on computers or mobile devices.
- Cloud hosting services are expected to provide backup storage and maintenance support.
- The design assumes DriverPass employees will receive training before using the platform.
- A third-part payment processor is assumed to securely manage customer payment transactions.

Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- The initial release may not support fully customizable training packages without additional development work.
- The system depends heavily on internet connectivity for real-time updates and synchronization.
- The initial deployment is limited to managing the company's current fleet of 10 vehicles and assigned instructors.
- Reservations are restricted to fixed two-hour lesson blocks based on current business requirements.
- Future feature additions or major system changes may require developer involvement.
- Budget, project deadlines, and staffing constraints may limit advanced functionality in the first release.

Gantt Chart

Please include a screenshot of the GANTT chart that you created with Lucidchart. Be sure to check that it meets the plan described by the characters in the interview.

